Beating the Bot: Best-Responding to LLM-Generated Strategies in a Five-Castle Colonel Blotto Contest

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Abstract

A tournament application asks entrants to split 100 soldiers across five castles worth 1–5 points, in two variants: a pairwise head-to-head game (Game 1) and a four-player placement game paying 9/4/2/0 (Game 2). Because many applicants are likely to paste the question into ChatGPT or Claude, the realized field may be dominated by large-language-model (LLM) answers. We query 8 current OpenAI and Anthropic models through their APIs using four prompt formats (full text, each game separately, one screenshot per game, both screenshots), collecting 167 Game 1 and 164 Game 2 allocations. LLM answers are highly clustered: in Game 1 the average answer is 3/10/22/30/35, almost every model puts the most soldiers on castle 5, and the single most common answer is the textbook "proportional" split 7/13/20/27/33. Best-responding to this field is easy. In Game 1, 5/16/0/36/43, which abandons castle 3 and outbids the cluster everywhere else, scores 11.50 of 15 against the LLM field (the LLM field itself averages 7.50). In Game 2, an allocation that takes castles 5, 2 and 1 (8 points, beating the 7 points available from castles 3 and 4) finishes first in almost every group, scoring 8.88 of a possible 9 versus a field average of 3.71. Stress tests show the Game 1 answer is robust to copycats while the Game 2 answer is fragile; we recommend a hedged Game 2 allocation, 9/21/0/0/70, that gives up 0.2 points against the pure LLM field but holds up far better if other applicants find the same exploit.

1 Introduction

Colonel Blotto games [1, 2] ask players to divide a fixed budget of resources over several battlefields, each won by whoever commits more. They have no pure-strategy equilibrium, so in practice the winning entry is the one that best anticipates what everyone else submits. Laboratory and online experiments find that people play Blotto with a few levels of iterated reasoning rather than equilibrium [3], and public Blotto contests, such as FiveThirtyEight's "Riddler Nation" castle tournaments [4], repeatedly show that the field is far from equilibrium and that a well-targeted best response to the previous round's entries wins.

This paper studies a version of the game that appears in a tournament application. Its premise is simple: in 2026, a large share of applicants will not reason about the game themselves. They will paste it into a chatbot and type in whatever comes back. If that is true, the field is predictable, because LLMs given the same prompt produce similar answers. We measure that predictability directly and compute allocations that exploit it.

Contributions.

1. A dataset of 167 + 164 allocations from 8 frontier and mid-tier LLMs (Section 3), with full transcripts.
2. A characterization of how LLMs play the game: what they share across vendors, how much they vary, and how the prompt format shifts their answers (Section 4).
3. Best responses to the LLM field, with robustness checks against bootstrap resampling, naive human strategies, and copycat entrants (Section 5).

2 The Game

Each player chooses $x \in \mathbb{Z}^5_{\geq 0}$ with $\sum_i x_i = 100$. Castle i is worth $v_i = i$ points, so 15 points are available.

Game 1 (head to head). Against an opponent y, a player earns $\sum_i v_i \left(\mathbf{1}[x_i > y_i] + \frac{1}{2}\mathbf{1}[x_i = y_i]\right)$. The score is the average over all other entrants. A field playing against itself averages exactly 7.5.

Game 2 (four-way battle). Players are placed in random groups of four. Each castle goes to the largest bid, with ties splitting the castle's points. Players are ranked by castle points and paid 9, 4, 2 and 0 for 1st to 4th; tied players split the payouts of the places they occupy. The score is the expected payout over random groups. The payouts sum to 15 per group, so the field average is 3.75, and the maximum is 9.

Scoring implementation. Game 1 is computed exactly. Game 2 is estimated by Monte Carlo over random 3-opponent draws from the field (4,000–8,000 groups per evaluation; standard errors are below 0.05 for the scores we report). Castle-point totals are compared as exact multiples of 1/12 to avoid floating-point rank errors. The code is in `scripts/blotto.py`.

3 Data Collection

Models. We queried four Anthropic models (`claude-fable-5-1`, `claude-opus-5-5`, `claude-opus-5-5` with adaptive thinking at high effort, and `claude-sonnet-5-5`) and four OpenAI models (`gpt-6.1-sol`, `gpt-6.1-terra`, `gpt-6-luna`, `gpt-5.5`). Every call used the provider's API with default sampling settings, no system prompt, and no tools.

Prompt formats. Applicants may present the question in different ways, so we used four formats:

- *Full text*: the complete question, both games, in one message.
- *Split text*: each game's text in a separate, fresh conversation.
- *One screenshot*: a screenshot of a single game from the application page plus "What should I put?", in a fresh conversation per game.
- *Both screenshots*: both screenshots in one message plus the same question.

The screenshots omit the intro text ("100 soldiers … whole numbers"), which mirrors what an applicant screenshotting the form would send. Each model was run five times per format, plus one pilot full-text run, for 246 successful replies (one call timed out).

Extraction. A small model (`claude-haiku-5-5`) read each reply and extracted the final recommended allocation as JSON. Replies covering a single game were extracted with a single-game prompt. All extracted allocations were validated to be non-negative integers summing to 100. Two replies (both `gpt-6-luna`, Game 2, split text) declined to give an allocation and instead asked for assumptions about opponents. One reply gave a single allocation for both games and was hand-coded. All transcripts are in `responses/transcripts/`.

4 How LLMs Play

4.1 Game 1

Table 1 and Figure 1 summarize Game 1. The answers form one tight cluster. Every model's mean allocation lies within a few soldiers of 3/10/22/30/35, the bids increase with castle value, and 97% of answers put the most soldiers on castle 5. Only 26% abandon castle 1, and almost none (1%) abandon castle 3. There are 119 distinct allocations among 167, but many differ by a soldier or two. The most common single answer, given 9 times, is the proportional split 7/13/20/27/33, which scores just 3.66 against the field. Claude's answers are more homogeneous than GPT's (per-castle standard deviations of 1.6–4.1 versus 3.2–6.6 soldiers).

The "Score" column gives each group's average score when its answers are entered into the full LLM field. The models that skip or nearly skip castles 1–2 and push the surplus into 3–5 (`gpt-6.1-sol`, `gpt-5.5`, `claude-sonnet-5-5`) do best, at about 8.8. Models that stay close to proportional (`gpt-6-luna`) do worst, at 5.2. The best single answer in the field is 0/0/24/34/42 (`claude-sonnet-5-5`), scoring 11.24.

Table 1: Game 1 mean allocation by model (soldiers per castle) and average head-to-head score of that model's answers against the full LLM field (max 15).

Model	n	C1	C2	C3	C4	C5	Score
`claude-fable-5-1`	21	2.8	10.8	22.9	30.2	33.3	7.79
`claude-opus-5-5`	21	2.2	11.9	21.8	29.1	35.1	7.23
`claude-opus-5-5-thinking`	20	2.5	12.1	21.6	29.1	34.8	7.07
`claude-sonnet-5-5`	21	0.7	7.0	24.4	31.3	36.6	8.74
`gpt-5.5`	21	1.7	8.1	21.6	30.7	37.9	8.79
`gpt-6-luna`	21	5.6	11.3	20.3	27.8	35.0	5.19
`gpt-6.1-sol`	21	2.4	7.1	21.9	30.2	38.4	8.83
`gpt-6.1-terra`	21	3.6	12.4	20.8	29.0	34.2	6.35
All Claude	83	2.0	10.4	22.7	29.9	35.0	7.71
All GPT	84	3.3	9.7	21.2	29.4	36.4	7.29
All models	167	2.7	10.1	21.9	29.7	35.7	7.50

4.2 Game 2

Game 2 answers are more varied (132 distinct among 164) and more concentrated on the top castles (Table 2). 57% skip castle 1, 37% skip both castles 1 and 2, and 17% skip castle 3. Many models reason that first place requires winning castles 4 and 5 outright; the most common answer is 0/0/0/47/53. `gpt-6.1-sol` plays this way consistently (mean 0/0/10/37/53), which makes it the best-scoring model in Game 2 (5.14 of 9). `claude-sonnet-5-5`, which keeps a Game-1-like shape, scores lowest (1.98).

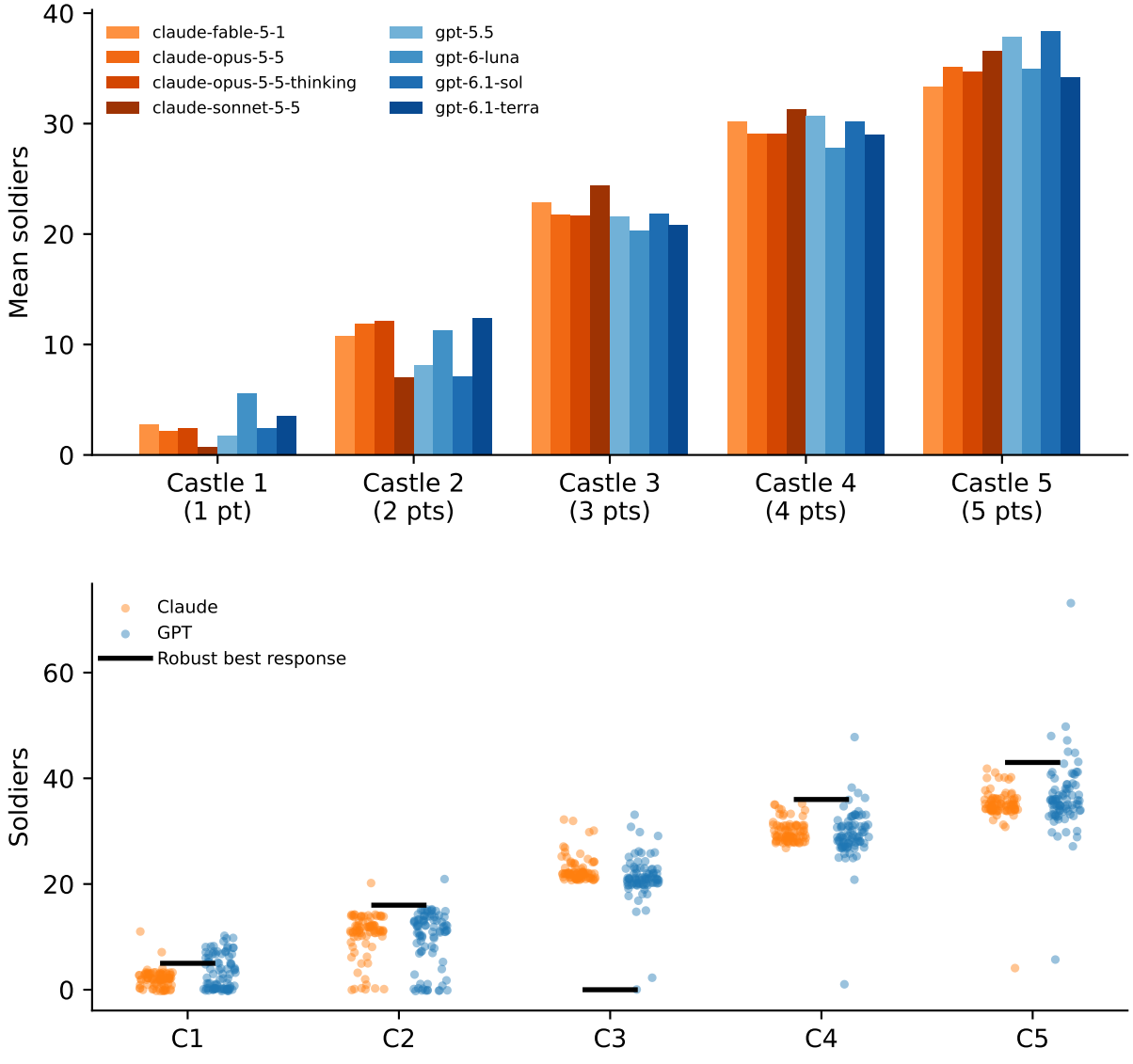


Figure 1: Game 1. Top: mean soldiers per castle by model (Claude in orange, GPT in blue). Bottom: every individual answer (jittered), with the robust best response marked in black.

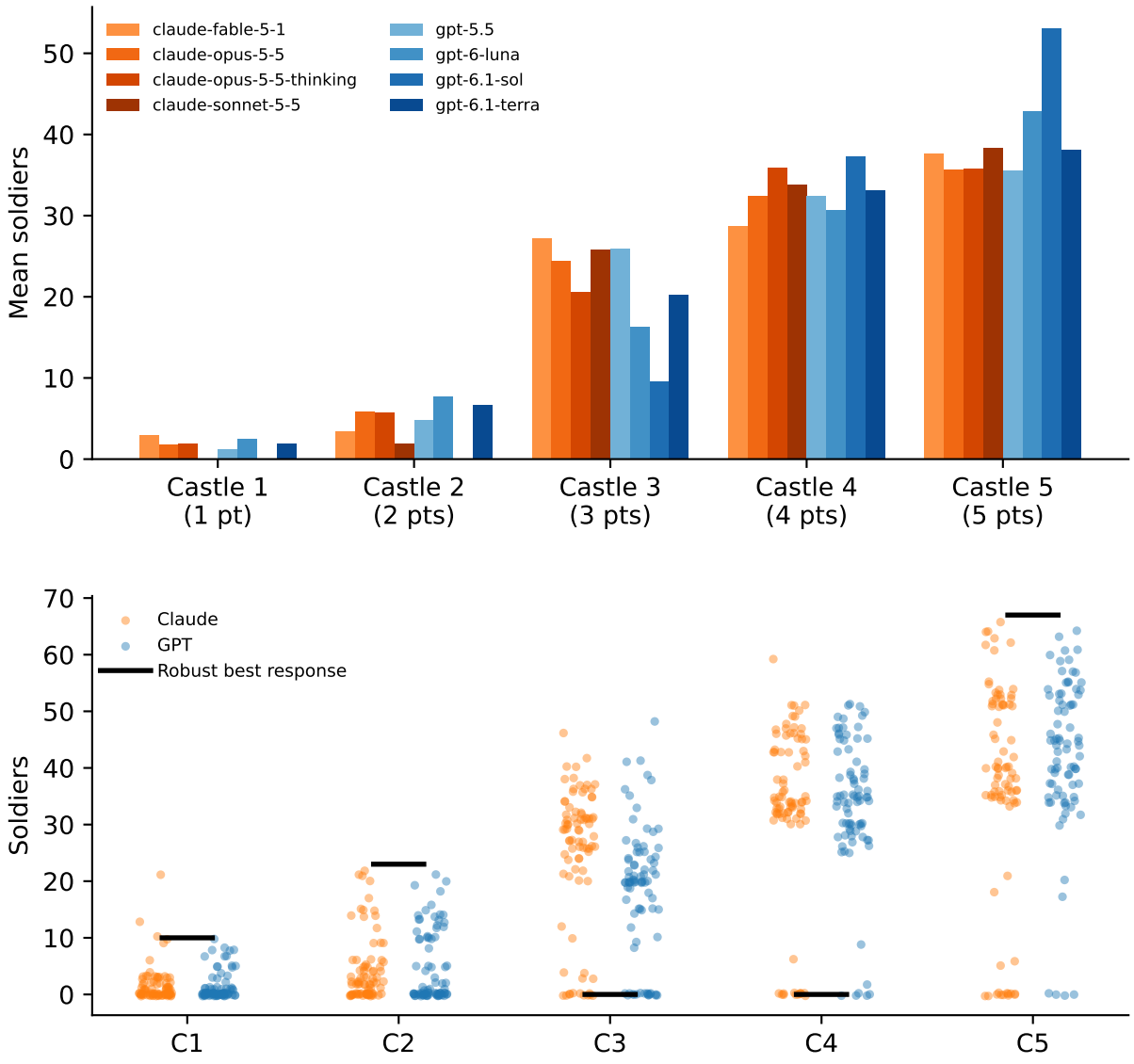


Figure 2: Game 2. Top: mean soldiers per castle by model. Bottom: individual answers with the best response marked in black.

Table 2: Game 2 mean allocation by model and the average expected payout of that model's answers in random four-player groups drawn from the full LLM field (max 9; payouts sum to 15 per group, so the field average is 3.75 up to Monte Carlo error).

Model	n	C1	C2	C3	C4	C5	Score
`claude-fable-5-1`	21	2.9	3.5	27.2	28.8	37.6	4.71
`claude-opus-5-5`	21	1.8	5.8	24.4	32.4	35.6	4.05
`claude-opus-5-5-thinking`	20	1.9	5.8	20.6	35.9	35.8	4.99
`claude-sonnet-5-5`	21	0.1	2.0	25.8	33.8	38.4	1.98
`gpt-5.5`	21	1.2	4.8	26.0	32.5	35.6	3.97
`gpt-6-luna`	19	2.5	7.7	16.3	30.7	42.9	2.54
`gpt-6.1-sol`	20	0.1	0.0	9.6	37.2	53.0	5.14
`gpt-6.1-terra`	21	1.9	6.6	20.2	33.1	38.1	2.29
All Claude	83	1.7	4.2	24.5	32.7	36.9	3.92
All GPT	81	1.4	4.8	18.1	33.4	42.3	3.49
All models	164	1.5	4.5	21.4	33.0	39.5	3.71

4.3 Effect of the prompt format

Table 3 breaks results down by prompt format. In Game 1 the format matters little. In Game 2 it matters a lot. When models see both games together (full text or both screenshots), they put about 46 and 43 soldiers on castle 5. When they see Game 2 alone (split text or one screenshot), it is about 35 and 32, with more spread over castles 1–3. Seeing the contrast with Game 1 seems to push models toward a more aggressive, placement-aware answer. Practically, the Game 2 field depends on how applicants paste the question, which is one more reason to prefer an allocation that does not depend on the field's exact shape.

Table 3: Mean allocation and score by prompt format. Top: Game 1. Bottom: Game 2.

Prompt method	n	C1	C2	C3	C4	C5	Score
Full text	47	2.4	8.7	23.0	30.3	35.6	8.07
Split text	40	2.5	10.1	22.1	28.8	36.5	7.53
One screenshot	40	3.5	11.4	20.9	29.2	35.0	6.85
Both screenshots	40	2.5	10.3	21.5	30.1	35.6	7.45

Prompt method	n	C1	C2	C3	C4	C5	Score
Full text	47	0.6	2.2	16.0	34.9	46.4	3.82
Split text	38	2.3	6.5	26.1	30.3	34.8	3.45
One screenshot	39	2.5	6.7	25.4	33.4	32.1	3.74
Both screenshots	40	1.1	3.1	19.3	33.2	43.2	3.79

4.4 What the models say

Replies are short. Claude models average about 350 words per reply and GPT models about 160. Only Claude names the game: 37 of 123 Claude replies mention "Colonel Blotto", against none of 123 GPT replies. Many replies across both vendors give the same advice: avoid round numbers to win ties (75 of 246 replies mention round numbers explicitly) and do not waste soldiers on castle 1. About 30 replies stress that there is no single correct answer. These shared heuristics explain the

clustering. Every model applies the same small adjustments ("a little more than proportional on the big castles, a few soldiers off the round numbers"), so they land in the same place.

5 Best Responses

Method. We find best responses by steepest-ascent hill climbing over integer allocations. Moves transfer $d \in \{1, 2, 3, 5, 8, 13\}$ soldiers between castles. Searches restart from the field mean, sampled field members, and random allocations. To guard against overfitting, we also compute best responses to bootstrap resamples of the field and choose, among all candidates, the one maximizing mean score plus half the 10th-percentile score across 20 bootstrap fields and an "AI + naive" field. That field adds simple human-style allocations (uniform, proportional, all-in on castles 4–5, skip castles 1–2, abandon castle 5), weighted to about 25% of entries. We call the result the *robust best response*.

5.1 Game 1

Table 4 reports scores for several strategies on different fields. The robust best response 5/16/0/36/43 scores 11.50 against the whole LLM field, 11.05 against GPT answers only, 11.95 against Claude answers only, and 10.88 with naive humans added. Across bootstrap fields its worst score is 10.88. The logic is clear from Figure 1. The LLM cluster puts about 22 soldiers on castle 3, about 30 on castle 4 and about 35 on castle 5. Giving up castle 3 (3 points) frees enough soldiers to beat nearly every LLM on castles 4 and 5 (9 points) and also on castles 1 and 2. Bidding 5 on castle 1 and 16 on castle 2 beats the 0–3 and 7–12 that LLMs typically send. The averaged GPT answer (3/10/21/29/37) scores only 7.87.

Table 4: Game 1: head-to-head score (max 15) of best responses and reference strategies against each opponent field. "BR(X)" is the hill-climb best response to field X.

Strategy	Allocation	All AI	Claude	GPT	AI + naive
BR(All AI)	5/16/0/36/43	11.50	11.95	11.05	10.88
BR(Claude)	5/15/0/36/44	11.47	11.95	10.99	10.86
BR(GPT)	1/0/24/34/41	11.29	11.48	11.11	10.85
Robust	5/16/0/36/43	11.50	11.95	11.05	10.88
Mean AI answer	3/10/22/30/35	7.83	7.85	7.81	7.98
Mean GPT answer	3/10/21/29/37	7.87	7.66	8.08	8.01
Proportional	7/13/20/27/33	3.66	2.81	4.49	4.19
Uniform	20/20/20/20/20	3.43	3.05	3.82	3.85

5.2 Game 2

Game 2 rewards placement, not points, so the exploit is structural (Table 5). Castles 1, 2 and 5 together are worth 8 points, and castles 3 and 4 together are worth 7. A player who wins castles 1, 2 and 5 outright therefore finishes ahead of anyone who wins only castles 3 and 4. The highest castle-5 bid in the LLM Game 2 field is 66, and 95% of answers send at most 8 soldiers to castle 1 and at most 17 to castle 2. The best response 10/23/0/0/67 wins those three castles in nearly every group and scores 8.88, close to the maximum of 9. The averaged LLM answer (2/4/21/33/40) scores only 1.88, worse than the even split.

Table 5: Game 2: expected placement payout (max 9) of best responses and reference strategies against each opponent field.

Strategy	Allocation	All AI	Claude	GPT	AI + naive
BR(All AI)	10/23/0/0/67	8.89	8.79	9.00	8.75
BR(Claude)	10/23/0/0/67	8.89	8.80	9.00	8.77
BR(GPT)	10/22/0/2/66	8.80	8.63	9.00	8.70
Robust	10/23/0/0/67	8.88	8.79	9.00	8.74
Mean AI answer	2/4/21/33/40	1.88	2.03	1.90	1.99
Mean GPT answer	1/5/18/33/43	2.06	2.36	1.72	2.17
Proportional	7/13/20/27/33	1.95	1.96	2.15	1.65
Uniform	20/20/20/20/20	2.43	2.24	2.79	2.29

5.3 Stress test: what if others find the exploit?

An applicant reading this paper, or simply thinking one level deeper, might play the same exploit. We add "exploiter" entries (the recommendation and small random perturbations of it) to the field at increasing shares (Table 6). The Game 1 recommendation degrades gracefully: with a quarter of the field playing the same idea it still scores 10.2. Matching allocations only tie, and ties split points. The Game 2 recommendation is brittle. Its 67 on castle 5 is exactly one more than the highest LLM bid, so any copycat with 67 or more on castle 5 takes that castle away, and with it first place. At 25% exploiters its score falls to about 5.8.

Table 6: Score of the robust best response, and the new best response, when exploiters make up a growing share of the field. Top: Game 1. Bottom: Game 2.

Share of exploiters	Recommendation score	New best response	Its score
0%	11.50	5/16/0/36/43	11.50
10%	10.99	3/15/1/37/44	11.11
25%	10.19	8/1/3/39/49	10.85
50%	8.88	8/1/3/39/49	11.56

Share of exploiters	Recommendation score	New best response	Its score
0%	8.89	10/23/0/0/67	8.90
10%	7.71	7/24/0/0/69	8.44
25%	5.81	14/28/5/52/1	8.39
50%	3.40	14/28/51/3/4	8.72

Table 7 compares hedged versions of the Game 2 exploit with more margin on castle 5. Moving to 9/21/0/0/70 costs about 0.2 points against the pure LLM field but keeps the score near 8.3 with 10% copycats and above 7.6 with 25% copycats. Going further (72 or 75 on castle 5) costs more against the LLM field and also starts to lose castles 1 and 2.

Table 7: Game 2 hedges: expected payout against the LLM field and against the field plus copycats (exploiters who bid 67–69 on castle 5).

Allocation	AI field	+10% copycats	+25% copycats
10/23/0/0/67	8.88	7.47	4.99
9/21/0/0/70	8.68	8.27	7.66
8/20/0/0/72	8.54	8.11	7.53
6/19/0/0/75	8.37	8.04	7.49
3/0/38/59/0	7.98	7.06	6.28
0/0/0/47/53	5.30	4.83	4.18

6 Recommendation

	C1	C2	C3	C4	C5
Game 1	5	16	0	36	43
Game 2	9	21	0	0	70

For Game 1 we recommend the robust best response 5/16/0/36/43. For Game 2 we recommend the hedged allocation 9/21/0/0/70 over the sharper 10/23/0/0/67, because the cost against the pure LLM field is small and the protection against copycats is large.

7 Limitations

- **API versus apps.** Most applicants use the ChatGPT and claude.ai apps, not the API. The apps add their own system prompts, may route to different models, and may use different sampling settings. Our field approximates the app field; it is not identical to it. Hand-collected app answers can be added to `responses/responses.csv` and the analysis rerun.
- **No real human data.** The "naive" entries are hypothetical. Real applicants who do not use an LLM, or who adjust the LLM's answer (for example by adding a few soldiers to castles 4 and 5), are not observed.
- **Other vendors.** We did not query Gemini, Grok, DeepSeek or other models applicants might use.
- **Strategic applicants.** Our best responses assume the field is mostly LLM answers. The stress test covers some deviation, but a field full of level-2 thinkers would call for a different answer.
- **Extraction.** Numbers were extracted by an LLM. Every extracted allocation was checked to sum to 100, and spot checks matched the transcripts, but the extractor could still occasionally pick an alternative allocation a reply mentions instead of its main pick.

8 Reproducibility

All code, prompts, screenshots, transcripts and data are in the repository. `scripts/collect.py` queries the models, `scripts/paper_stats.py` and `scripts/stress.py` generate every table, figure and number in this paper (fixed random seeds), and `tectonic paper/paper.tex` builds the PDF.

References

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